Optimal biomass-based climate change mitigation pathways are spatially heterogeneous due to localized technoeconomic and biophysical constraints

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2026-08-27

Abstract

The use of biomass for climate change mitigation offers substantial technical potential, but optimal deployment is highly dependent on spatial, technoeconomic, and biophysical constraints. Blanket climate policies frequently fail to capture this heterogeneity. Here we apply a spatially explicit techno-economic and consequential life cycle assessment framework deployed at a 0.1-degree resolution across the United States, Europe, India, and China. We evaluate the economic viability and carbon abatement potential of three distinct biomass utilization pathways: Bioenergy (BES), Bioenergy with Carbon Capture and Storage (BECCS), and Bioenergy with Biochar Carbon Storage (BEBCS). Localized constraints, such as distance to geological CO_2 sinks, baseline grid carbon intensity, and native soil cation exchange capacity, dictate the economically optimal pathway. Traditional BES tends to dominate at low carbon prices, particularly in regions with high wholesale energy prices like Europe, or high grid carbon intensities like India. Biochar (BEBCS) emerges as a highly competitive transition technology at moderate carbon prices, and serves as the optimal long-term pathway in specific agro-ecological zones with degraded soils, such as southern China. Conversely, high capital costs and piecewise long-distance pipeline penalties act as severe barriers to BECCS, which only achieves universal dominance at high carbon prices (e.g., \$150/tCO_2e) where carbon capture revenues finally overwhelm infrastructure penalties. Ultimately, effective climate mitigation requires localized policies that match biomass resources to their most efficient regional end-use.

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1 Introduction

Biochar has been widely promoted for climate change mitigation, with co-benefits for soil health and crop production when used as a soil amendment.(1, 2) Globally, biochar has the sustainable technical potential for carbon sequestration and emissions reductions of up to 3.4–6.3 Pg CO_2e. However, the costs, benefits and tradeoffs for biochar vary spatially, creating substantial heterogeneity in its feasibility at a given carbon price. This variability is further affected by the fact that biochar must compete with alternative uses for the limited supply of available waste biomass. Indeed, alternative uses of biomass, such as producing bioenergy with or without carbon capture and sequestration can also help to mitigate climate change and may provide greater climate mitigation, lower costs, and/or greater co-benefits in certain contexts.(3) This depends on a number of different factors, including soil type, cropland productivity, climate, energy fuel mix, feedstock availability, distance to CO_2 sinks for carbon capture and storage, energy prices and food prices. For example, bioenergy production may provide a greater climate benefit in regions that are highly dependent on carbon-intensive energy sources and have high soil fertility, whereas biochar is more favorable where low-fertility soils can benefit from its soil-health improvement.(3)

It is, therefore, critical to understand where biochar could be the best use of resources from a climate mitigation perspective. Currently, most biochar assessments focus on the technical potential, without mapping out the tradeoffs or identifying optimal pathways based on location. While there are some biochar life-cycle assessments that account for alternative biomass uses, they are highly context-specific and difficult to scale up on(4, 5). As such, this project aims to provide a detailed feasibility and tradeoff analysis across four regions (North America, India, China, and Europe) to map where biochar represents the most efficient use of biomass to reduce atmospheric greenhouse gasses (GHGs). These regions represent a diversity of geographic, socio-economic, and agro-ecological conditions for a broad assessment of how constraints, tradeoffs and benefits vary across regions. This work will allow policy-makers and practitioners to make scientifically-informed decisions about which mitigation pathways are best suited for their geographies. It can also help contextualize biochar's mitigation potential by identifying areas that maximize its climate benefits and realistically support its production and application.

2 Methods

2.1 Model Architecture

The C-SCAPE (Carbon Spatial Cost and Allocation Pathways Evaluator) model (version 1.0) evaluates three biomass utilization pathways: Bioenergy (BE), Bioenergy with Carbon Capture and Storage (BECCS), and

Bioenergy with Biochar Carbon Storage (BEBCS). The model evaluates a spatially-varying net present value (NPV) for each technology using a discounted cost-benefit analysis, with GHG emissions and removals converted to 100-year CO_2 equivalent values and internalized into the economic analysis using an assumed carbon price.

The model was deployed across the coterminous United States, Europe, India, and China. All spatial inputs were projected into region-specific, equal-area coordinate reference systems (EPSG:5070, EPSG:3035, EPSG:8859, and ESRI:102025, for the US, Europe, India, and China, respectively), and harmonized to a 0.1-degree resolution. The model was implemented in the R programming language, using the Terra package for spatial analysis.

The following model description outlines the assumptions,m equations and parameters used in the model. The default parameter values are provided in a table at the end of the Methods.

2.1.1 Biomass Feedstock Sourcing and Logistics

2.1.1.1 Biomass Transport Costs and Emissions

High resolution rasters of available crop residue biomass were derived from Karan et al.(6). The average transport distance for biomass was calculated from this using focal statistics based on the annual biomass requirement for the conversion facility. The required annual biomass mass (M_{req}) is determined by the target thermal capacity, the capacity factor, and the biomass lower heating value (LHV). A spatial focal sum algorithm was applied to aggregate the surrounding biomass over a discrete series of expanding radii (from 5 km up to 500 km). The required collection radius was then calculated via quadratic interpolation between these radial bins. All domains were uniformly projected and resampled to the 0.1 degree ($\sim$ 10 km) equal-area grid prior to executing the focal sum algorithms to ensure methodological consistency. Assuming a circular catchment geometry, the average straight-line collection distance is approximated as two-thirds of the interpolated bounding radius (i.e. assuming homogeneous distribution within the catchment area). The average straight-line distance was multiplied by a tortuosity factor of 1.3 to calculate the effective road transport distance.(7) Biomass logistical costs were modeled using a fixed material handling and loading cost and a variable trucking cost applied to the effective road distance. Scope three transportation emissions associated with feedstock procurement were estimated using a heavy-duty diesel transport emissions factor applied to the effective distance and deducted from the net carbon abatement.

2.1.1.2 Fuel Quality

Agricultural crop residues typically have a higher ash content than the wood chips more commonly used in bioenergy combustion. High-ash feedstocks carry significant fouling and slagging risks, requiring additional capital expenditures (CAPEX) and operating expenditure (OPEX), for example through deployment of Circulating Fluidized Bed (CFB) boilers. This fuel quality penalty is modelled as a proportional (25%) increase in base CAPEX for combustion technologies (BE and BECCS), a 50% increase in operations and maintenance (O&M) costs, and a 10% reduction in overall electrical conversion efficiency, relative to a wood-chip fired bioenergy plant.(*8*–*10*)

2.1.1.3 Regional Pricing Structures

Feedstock pricing is not treated as a static global constant; rather, it is derived via a spatially explicit shadow-pricing logic converted to a normalized USD/Mg equivalent.

- **United States:** Establishes a baseline farm-gate cost ($70.00/Mg) combined with a nutrient replacement penalty (e.g., $25.06/Mg for corn stover) to compensate for the export of field nutrients.
- **Europe:** Utilizes a baseline NUTS-3 regional roadside cost, modified by a temporal volatility multiplier that accounts for interim storage costs (up to a 36.4% premium for 6-month storage).
- **India:** Capitalizes on the necessity to avoid crop stubble burning. The model assumes a $0 stumpage fee, but imposes a step-cost for densification: raw bale transport costs apply under 50 km, while more expensive pelletization costs are triggered for catchment zones exceeding 50 km.
- **China:** Modifies a baseline plant-gate cost with exogenous spatial risk multipliers, applying premiums for localized weather volatility and rural expansion risks.

2.1.2 Technology Pathways and Thermodynamics

The core thermodynamic engine evaluates three distinct biomass utilization pathways, processing the standardized 250 MW_{th} biomass input into varying ratios of exported electrical energy and sequestered carbon.

2.1.2.1 Bioenergy (BES)

The BES pathway represents a traditional, dedicated biomass combustion facility. In this baseline scenario, the feedstock is fully oxidized to maximize electricity exported to the regional grid. Because there is no active carbon capture, the physical carbon embedded in the biomass is returned to the atmosphere.

The technoeconomic parameters for this modern combustion archetype assume an electrical conversion efficiency of 30%. Capital expenditures (CAPEX) are standardized at $\$3,000/\text{kW}_e$, scaling non-linearly with plant capacity via a standard 0.7 scaling factor. Operations and maintenance (O&M) are assessed as a fixed annual rate of 4% of total CAPEX. Total electrical output is derived directly from the biomass lower heating value (LHV, assumed as 18.6 GJ/Mg for dry, ash-free feedstock) and the plant's operational capacity factor (85%).

2.1.2.2 Bioenergy with Carbon Capture and Storage (BECCS)

The BECCS pathway augments the standard combustion baseline with a post-combustion carbon capture unit. The addition of this unit imposes a significant parasitic energy penalty, primarily driven by solvent regeneration and CO_2 compression requirements. Consequently, the net electrical conversion efficiency drops from the 30% BES baseline to 28%.

The capture process is parameterized with a 90% capture efficiency rate. The total volume of captured CO_2 routed to geological storage is calculated stoichiometrically based on the carbon fraction of the incoming biomass feed (default 48% C), converted to CO_2 mass equivalents using the 44/12 molar ratio. The increased infrastructure complexity elevates the base CAPEX to $\$4,000/\mathrm{kW}_e$, with annual O&M scaling concurrently to 5% of CAPEX.

2.1.2.3 Bioenergy with Biochar Storage (BEBCS) and Pyrolysis Physics

Unlike the fully oxidative pathways, the BEBCS pathway relies on thermochemical pyrolysis, producing a solid, recalcitrant carbon matrix (biochar) alongside volatile syngas and bio-oil. The model employs a rigorous mass and energy balance approach constrained by elemental conservation of Carbon, Hydrogen, and Oxygen (C, H, O).

The mass yield and elemental composition of the resulting biochar are determined dynamically as a function of the peak pyrolysis temperature (T_{py}) agarnd the lignin fraction of the feedstock. Specific yields for non-condensable gases—hydrogen (H_2), carbon monoxide (CO), methane (CH_4), and ethylene (C_2H_4)—are calculated explicitly based on temperature-dependent reaction kinetics.

To close the mass balance, the unaccounted elemental fractions (U_c, U_h, U_o) are stoichiometrically partitioned into bio-oil, water (H_2O), and carbon dioxide (CO_2). The model restricts bio-oil to a fixed empirical composition (62.5% C, 7.56% H, 29.94% O) to solve the final product distribution analytically.

The net usable energy flux for the BEBCS pathway is calculated by subtracting the process heat requirements from the total enthalpy of the gaseous and liquid products. The higher heating value (HHV) of the biochar is calculated via the Dulong formula, utilizing the temperature-dependent C, H, and O fractions. Heat for the endothermic pyrolysis reaction is supplied entirely by the combustion of the volatile syngas and bio-oil fractions; any surplus thermal energy is converted to electricity using the baseline BES efficiency parameters. If the syngas and bio-oil are insufficient to meet the thermal load, a portion of the biochar is oxidized to close the energy balance, proportionally reducing the final carbon sequestration yield.

2.1.3 Carbon Transport and Geological Storage

2.1.3.1 Spatial Routing and Terrain Friction

The spatial routing of captured CO_2 is governed by an economic optimization algorithm that dynamically selects the lowest-cost transport modality—pipeline or maritime shipping—for every grid cell. Standard spatial optimization models frequently rely on simple Euclidean distances or Voronoi tessellations, which artificially generate circular geographical discontinuities in net present value (NPV) maps. To eliminate these artifacts and accurately model the economic constraints of CO_2 transport, this framework implements a deterministic Lowest Cost Routing logic that relies on a terrain-optimized Least Cost Path (LCP) algorithm rather than linear assumptions.

Topographic friction surfaces were generated by deriving the slope (in degrees) from the Global Digital Elevation Model (GDEM). To preserve localized linear barriers—such as mountain ridges or steep ravines—during spatial aggregation to the base model resolution, the high-resolution slope data was aggregated using a 95th percentile percolation threshold. The terrain friction surface was then parameterized using an exponential cost function:

$$M(\theta) = \exp(0.25 \cdot \theta)$$

where θ represents the aggregated slope. Furthermore, to ensure that proposed CO_2 infrastructure avoids ecologically sensitive or legally restricted regions, protected areas from the World Database on Protected Areas (WDPA) were rasterized and treated as absolute barriers. This forces the routing algorithm to find the most efficient path around protected zones. Once the minimal effective transport route is mapped, the distances are adjusted by a standard tortuosity multiplier of 1.25 to account for right-of-way routing and localized physical avoidance.

2.1.3.2 Maritime Shipping Modality

For offshore geological sinks, or for highly isolated onshore sources, CO_2 is transported via maritime or rail networks. This modality is modeled with a step-cost function dominated by the capital-intensive fixed entry costs of liquefaction and terminal buffering, paired with highly efficient variable voyage costs:

$$C_{shipping} = C_{liquefaction} + C_{terminal} + (c_{voyage} \cdot d_{effective})$$

2.1.3.3 Hub-and-Spoke Pipeline Modality (Piecewise Linear)

Standard length-invariant TEA cost functions for CO_2 pipelines adequately model regional networks but fundamentally underestimate the capital and energetic burdens of long-distance transport. Specifically, maintaining CO_2 in a dense or supercritical phase over extended distances requires the deployment of intermediate booster compression stations to overcome frictional pressure drops.

To simulate this without computationally intensive grid-level pump topology, pipeline CAPEX is calculated using a piecewise linear hub-and-spoke framework:

1. **The Feeder Leg:** Sources build a dedicated feeder pipeline for the first 50 km (d_{feeder}), bearing the full capacity-scaled CAPEX.
2. **The Trunk Leg (Base Phase):** For distances up to 700 km, the source connects to a regional trunkline operating at an efficient scale (e.g., 3 Mtpa), paying only a proportional mass share of the CAPEX.
3. **The Trunk Leg (Booster Phase):** For distances exceeding 700 km, the marginal cost of the trunkline increases by a penalty factor ($P_{booster} = 2.0$) to approximate the compounding capital and operational costs of intermediate booster stations.

The trunkline CAPEX share ($CAPEX_{trunk}$) is calculated as:

$$CAPEX_{trunk} = \begin{cases} C_{base} \cdot (d_{effective} - 50) & \text{if } d_{effective} \leq 700 \\ C_{base} \cdot (700 - 50) + (C_{base} \cdot P_{booster}) \cdot (d_{effective} - 700) & \text{if } d_{effective} > 700 \end{cases}$$

where C_{base} is the unit length cost of the efficient-scale trunkline, scaled to the site's proportional mass.

2.1.3.4 Modality Optimization

To eliminate artifactual spatial boundaries, the algorithm evaluates both modalities for all onshore sinks and selects the optimal route:

$$Cost_{transport} = \min(C_{pipeline}, C_{shipping})$$

Because of the piecewise booster penalty, the pipeline cost curve steepens significantly beyond 700 km. This ensures the pipeline cost curve intersects with the flat $C_{shipping}$ cost floor, allowing the model to smoothly transition to shipping modalities at very long distances.

2.1.3.5 Biochar Permanence and Agronomic Valuation (BEBCS Feedbacks)

The Bioenergy with Biochar Storage (BEBCS) pathway diverts a fraction of the original biomass carbon into a recalcitrant solid matrix. While physical disposal methods such as the landfilling of biochar represent a highly effective method to provide highly stable carbon storage, this framework strictly evaluates soil application to quantify the cascading agronomic co-benefits and regional substitution values. Consequently, the carbon retention within the soil matrix is finite and must be modeled dynamically based on localized biogeochemical conditions.

2.1.3.5.1 Biochar Carbon Permanence (F_{perm})

To accurately quantify the net carbon dioxide removal (CDR) of the BEBCS pathway, the model calculates the permanent carbon fraction (F_{perm}) that remains stabilized in the soil after a 100-year time horizon. This fraction is derived using the multi-pool decay methodology established by (*11*).

The baseline degradation rates of the biochar's labile and recalcitrant carbon pools are fitted to either the peak pyrolysis temperature (T_{py}) or the molar Hydrogen-to-Organic-Carbon (H:C) ratio of the final char. Because microbial and chemical degradation kinetics are highly temperature-dependent, these reference decay rates are spatially adjusted to match the localized mean annual soil temperature using a Q_{10} temperature sensitivity function. To optimize spatial computational efficiency across the high-resolution rasters, the thermodynamic module utilizes bilinear interpolation over a pre-calculated matrix of F_{perm} values bounded by soil temperatures (-55°C to 40°C) and H:C ratios (0.0 to 0.7).

Total carbon sequestered per megagram of feedstock is therefore modeled as:

$$C_{seq} = Yield_{bc} \cdot C_{content} \cdot F_{perm} \cdot \left(\frac{44}{12}\right)$$

where $Yield_{bc}$ is the mass fraction of biochar produced, $C_{content}$ is the carbon mass fraction of the biochar, and the 44/12 ratio converts the elemental carbon to CO_2 mass equivalents.

2.1.3.6 Mechanistic Agronomic Valuation

Previous spatial models frequently value biochar using a static, exogenous global market price. To better reflect real-world agricultural economics, C-SCAPE endogenously determines the economic credit of biochar through a mechanistic substitution valuation. This assumes that the rational economic value of biochar is equal to the cost of the conventional agricultural inputs it successfully offsets:

$$Value_{ag} = Value_{lime} + Value_{nutrients} + Value_{physical}$$

1. **Liming Equivalence ($Value_{lime}$):** The model spatially evaluates the native soil pH against a neutral agricultural target (pH 6.5). If the soil is acidic, the biochar's Calcium Carbonate Equivalent (CCE, default 15%) is credited against the localized market price of bulk agricultural lime (e.g., \$60/Mg). If the regional soil is already alkaline, the liming value is strictly zero.
2. **Nutrient Substitution ($Value_{nutrients}$):** The available nitrogen (N), phosphorus (P), and potassium (K) fractions within the biochar are credited against prevailing regional fertilizer costs. Baseline valuations rely on standard commercial substitutes, deriving elemental prices from Urea (\$0.92/kg N), Diammonium Phosphate (\$1.10/kg P_2O_5), and Potash (\$0.62/kg K_2O).
3. **Physical/CEC Improvements ($Value_{physical}$):** Biochar provides long-term yield efficiency gains by increasing water retention and buffering nutrient leaching, effects that are most pronounced in degraded or sandy soils. This value is heuristically scaled inversely to the soil's native Cation Exchange Capacity (CEC). Soils with severe CEC deficits receive higher annual premiums (up to \$50/Mg), while heavy clay soils (CEC $>$ 30 cmol/kg) receive zero premium.

Because physical and chemical soil improvements degrade or saturate over time, the total annual physical and nutrient benefits are amortized as a Present Value of an Annuity over a conservative 10-year agronomic impact duration, discounted at the baseline financial discount rate.

2.1.3.7 Consequential Life Cycle Assessment and Climate Mitigation

To robustly account for the climate mitigation benefit of bioenergy generation, the framework evaluates greenhouse gas offsets using a consequential Life Cycle Assessment (LCA) boundary. Total carbon abatement is calculated distinctively for each technology pathway, balancing active sequestration, grid displacement credits, and Scope 3 supply chain emissions.

2.1.3.7.1 Net Carbon Abatement Balances

The net carbon abatement for each grid cell is defined as the sum of physical carbon sequestered and emissions displaced, minus the transportation emissions penalty:

- **BES:** Abatement relies entirely on the displacement of grid emissions, minus the Scope 3 emissions associated with diesel trucking of the biomass feedstock.
- **BECCS:** Abatement includes the physical CO_2 captured and injected into the geological sink (assuming a 90% capture rate) plus the grid displacement credit. However, because the parasitic energy load of the capture process reduces the net electrical efficiency (from 30% to 28%), the total energy exported—and thus the resulting grid displacement credit—is proportionally lower for BECCS than for pure BES[^5].
- **BEBCS:** Abatement combines the grid displacement credit with the permanent CO_2 equivalent of the biochar carbon retained in the soil (F_{perm}), alongside a minor fixed soil greenhouse gas abatement credit (e.g., suppressed N_2O emissions).

For all pathways, Scope 3 feedstock transport emissions are modeled dynamically by applying a heavy-duty diesel transport emissions factor (0.0001 Mg CO_2e per Mg $\cdot$ km) to the effective biomass collection distance.

2.1.3.8 Grid Displacement Methodology (Marginal Carbon Intensity)

We estimate the Marginal Carbon Intensity (MCI) of displaced energy under two bounding scenarios to reflect differing macro-economic climate trajectories.

2.1.3.9 Scenario A: Empirical Baseline (Short-Run Marginal Capacity)

Under a Business-as-Usual (BAU) trajectory, the model assumes that new bioenergy capacity displaces the technology mix currently expanding on regional grids. Using multi-country historical electricity supply data (e.g., Ember, EIA, Eurostat), the model isolates generation sources experiencing positive net growth ($\Delta G > 0$) over a trailing five-year window (2018–2023).

The regional MCI is calculated as a generation-weighted average of these expanding sources:

$$MCI = \frac{\sum(\Delta G_i \times CI_i)}{\sum \Delta G_i}$$

where ΔG_i represents the net positive generation added for source i, and CI_i represents the static Intergovernmental Panel on Climate Change (IPCC) lifecycle greenhouse gas intensity for that specific technology. Standard IPCC reference intensities utilized include coal (820 gCO_2eq/kWh), natural gas (490 gCO_2eq/kWh), dedicated biomass (230 gCO_2eq/kWh), utility solar (48 gCO_2eq/kWh), wind (11.5 gCO_2eq/kWh), and nuclear (12 gCO_2eq/kWh). Applying this short-run marginal logic yields empirical baseline MCIs of approximately 278 gCO_2eq/kWh for the United States and 528 gCO_2eq/kWh for China.

2.1.3.10 Scenario B: Paris-Aligned Counterfactual (Long-Run Structural Shift)

To account for structural shifts in energy systems operating under strict climate stabilization limits (e.g., 1.5°C or 2.0°C targets), a secondary counterfactual scenario is evaluated. In a deep mitigation regime, fossil fuels are systematically phased out by hard policy caps. Therefore, the deployment of a baseload negative-emissions technology like BECCS or BEBCS does not offset high-emission fossil generation; rather, it displaces or delays the deployment of the next most capital-intensive low-carbon baseload alternative.

Under this sensitivity mode, the model assumes the displaced marginal generation is nuclear power. This scenario aggressively compresses net mitigation margins, dropping the grid displacement credit to a standardized lifecycle factor of 12 gCO_2eq/kWh globally.

2.1.4 Model Parameters and Data Sources

To ensure full reproducibility and transparency, the operational, financial, and biophysical parameters utilized within the C-SCAPE framework are tabulated below. Parameters are classified by their spatial resolution: **Scalar** values are held constant globally; **Regional** values act as constants but are overridden based on the macro-economic domain (e.g., India, China, Europe, US); and **Spatial** values are resolved dynamically at the grid-cell level using high-resolution raster datasets.

Table 1: Operational, financial, and biophysical parameters utilized within the BiocharAG framework.

		Value				
Parameter	Description	USA/Default	Europe	China	India	Units
Financial & Market						
`discount_rate`	Cost of capital	0.05	0.045	0.045	0.1	Fraction
`elec_price`	Wholesale electricity price	Spatially explicit				USD / MWh
`wholesale_discount`	Ratio of wholesale to retail price	0.4	0.4	0.4	1	Fraction
`c_price`	Carbon abatement credit price	50				USD / tCO_2e
`bes_life` / `beccs_life`	Infrastructure project lifetime	30				Years
`capacity_factor`	Annual plant capacity utilization	0.85				Fraction
`scaling_factor`	Capital cost scale exponent	0.7				—
Technology & Thermodynamics						
`plant_mw_th`	Target plant thermal capacity	250				MWth
`bes_capital_cost`	Base CAPEX for BES	3750	3450	2100	1950	USD / kWe

Table 1: Operational, financial, and biophysical parameters utilized within the BiocharAG framework.

		Value				
Parameter	Description	USA/Default	Europe	China	India	Units
`beccs_capital_cost`	Base CAPEX for BECCS	5000	4600	2800	2600	USD / kWe
`bes_om_factor`	O&M cost as fraction of CAPEX	0.045	0.045	0.03	0.025	Fraction
`beccs_om_factor`	O&M cost for BECCS	0.055	0.055	0.035	0.03	Fraction
`bes_energy_efficiency`	Electrical efficiency for BES	0.3				Fraction
`beccs_efficiency`	Electrical efficiency for BECCS	0.28				Fraction
`capture_rate`	BECCS CO2 capture efficiency	0.9				Fraction
`py_temp`	Peak pyrolysis temperature	500				°C
`heat_loss`	Pyrolysis process heat loss	2.35				GJ / Mg
`bo_hhv`	Bio-oil higher heating value	27.63				GJ / Mg
`bo_c`, `bo_h`, `bo_o`	Bio-oil elemental mass fractions	0.625, 0.0756, 0.2994				Fraction
Feedstock & Logistics						
`bm_lhv`	Biomass lower heating value	18.6				GJ / Mg
`bm_c`	Biomass carbon mass fraction	0.48				Fraction
`bm_ash`	Biomass ash mass fraction	0.05	0.05	0.05	0.15	Fraction
`bm_transport_fixed`	Fixed loading/handling cost	5				USD / Mg
`bm_transport_var`	Variable trucking cost	0.15				USD / Mg / km
`transport_EF`	Heavy-duty truck emissions	0.0001				Mg CO_2e / Mg · km
CO~2~ Transport Infrastructure						
`ccs_storage_cost`	Onshore injection and monitoring	12				USD / tCO_2
`cost_offshore_storage`	Offshore injection and monitoring	40				USD / tCO_2
`feeder_threshold`	Dedicated feeder pipeline limit	50				km
`booster_threshold`	Distance triggering booster penalty	700				km
`booster_penalty`	Marginal cost multiplier for boosters	2				—
`trunk_mass_flow`	Ref. capacity for trunkline sharing	3000000				Mg / year
`liq_cost`	CO_2 Liquefaction cost (shipping)	20				USD / tCO_2
`term_cost`	Terminal handling cost (shipping)	15				USD / tCO_2

Table 1: Operational, financial, and biophysical parameters utilized within the BiocharAG framework.

		Value				
Parameter	Description	USA/Default	Europe	China	India	Units
voyage_rate	Variable voyage cost (shipping)	0.035				USD / tCO_2 / km
transport_elec_price	Electricity price for compression	0.1				USD / kWh
Biochar & Agronomic						
h_c_org	Molar H:C ratio for permanence	0.35				Molar Ratio
time_frame	Time horizon of pyC permanence	100				Years
target_ph	Target agricultural soil pH	6.5				pH
bc_cce	Calcium Carbonate Equivalent	0.15				Fraction
price_lime	Market price for agricultural lime	45	50	35	35	USD / Mg
price_n	Market price for Nitrogen	1.59	1.75	0.79	0.14	USD / kg N
price_p	Market price for Phosphorus	2.08	2.29	1.1	0.7	USD / kg P_2O_5
price_k	Market price for Potassium	0.82	0.9	0.55	0.68	USD / kg K_2O
avail-n / -p / -k	First-year nutrient availability	0.1, 0.5, 0.8				Fraction
ag_impact_duration	Duration of nutrient/liming benefit	5				Years
soil_ghg_abatement	Suppressed soil emissions credit	0.1				tCO_2e / Mg
Spatial & Environmental						
ff_c_intensity	Displaced grid carbon intensity	Spatially explicit				tCO_2eq / GJ
biomass_density	Available spatial crop residues	Spatially explicit				Mg / km^2
soil_ph	Native topsoil pH	Spatially explicit				pH
soil_cec	Cation Exchange Capacity	Spatially explicit				cmol / kg
soil_temp	Mean annual soil temperature	Spatially explicit				°C

3 Results

3.1 Capital Intensive Pathways are Highly Sensitive to Discount Rates and Carbon Prices

Figures fig-evap-usa to fig-evap-china illustrate the core vulnerability of CDR pathways to capital lock-out. Because both BECCS (heavy upfront infrastructure) and BEBCS (deferred agronomic value) are highly sensitive to the cost of capital, high discount rates act as a strong barrier to adoption, forcing a retreat to traditional Bioenergy (BES) without high carbon prices. For example, in the USA, at a 2% social discount rate and a $30/t carbon price, BEBCS covers a large extent of the US Midwest and South, with BECCS only being viable close to CO_2 sinks. However, stepping up to the 8% market baseline eliminates BEBCS's advantage, replacing it with almost entirely with BES. The value of biochar relies heavily on liming and nutrient substitution benefits that are amortized over a 10-year agronomic impact duration. At an 8% or 15% discount rate, the present value of those future soil benefits is diminished, rendering the upfront thermal penalties of pyrolysis economically unviable at low carbon prices. At a 2% discount rate and intermediate carbon price (around $100/t), BECCS is nearly universal across the US. At 15%, the BECCS dominance is fractured, with BES reclaiming the geographic margins (e.g., the Northeast and far West), and BEBCS remaining only in localized pockets where soil conditions make it particularly attractive. This occurs because, when capital is expensive, the penalty for long-distance pipelines becomes insurmountable until carbon prices reach the $150/t range. By the time the carbon price reaches $150/t, the maps become almost entirely red across all discount rates. At this pricing level, the high CDR capture efficiency overwhelms the high cost of capital, making BECCS the most economically attractive option.

3.2 Biochar provides a mid-price transition technology option and a long term solution for specific agro-ecological zones

Figure fig-evap-china provides a counterpoint to the US results. It shows how soils with strong production constraints that can be alleviated by biochar can create a persistent competitive advantage for biochar, particularly when combined with large distances to CO_2 storage sites. BEBCS performs a central role as a transition technology that peaks at intermediate carbon prices that are high enough to make its carbon sequestration competitive with BES, but not so high that the more efficient CDR of BECCS per unit biomass outcompetes it (Figures fig-MACC-eur, fig-MACC-india, and fig-MACC-china). Nonetheless, while BECCS ultimately becomes the optimal technology across the entire US spatial domain at high carbon prices, BEBCS remains the most viable pathway in southern China across almost all price and discount rate scenarios. This demonstrates that BEBCS is not merely a transition technology, but the optimal long-term pathway for specific agro-ecological zones. In Southern China, characterized by highly weathered, acidic soils, the biochar's liming and nutrient use efficiency benefits provide a substantial economic credit that buffers against high processing costs. The model scales physical yield efficiency gains inversely to the soil's native Cation Exchange Capacity (CEC), thus granting severe CEC deficits higher premiums. This is confirmed by the SHAP analysis (Figure fig-shap-dominant), which identifies soil CEC as the dominant spatial driver in this region.

3.3 Regional Policies Hinder Biochar Competitiveness

In India, on the other hand, the mid carbon price advantage of BEBCS is almost entirely eliminated, with BECCS eventually covering most of the map at high carbon prices. Here, fertilizer subsidies hamper BEBCS's economic viability. Because BEBCS economics relies on improved nutrient use efficiency and lower fertilizer inputs per unit crop yield, cheap subsidized synthetic fertilizers erode biochar's nutrient substitution value. Despite this, Biochar maintains a strong foothold in the regions in the North East, North West, and central Eastern coastal regions of India where low soil CEC is a major constraint to crop production.

3.4 Regional Energy prices and Carbon Intensity provide a critical constraint on the optimal CDR pathways

At lower carbon prices The optimal technology in Europe is almost universally BES, driven by high energy prices as the dominant driver across Europe. At \$30 / t CO_2, the revenue from maximized power generation in BES outweighs the incentive to capture and sequester carbon, rendering the capital expenditure for BECCS and BEBCS unviable across all discount rates. As carbon prices increase, BECCS begins to dominate across central and northern Europe. BEBCS gains a strong advantage in Southern Europe, and Eastern Scandinavia where various combinations of large distances to CO_2 sinks, already-low carbon intensity of the energy systems, and infertile soils make it more competitive. As in China, we see BEBCS maintain its advantage in these regions, even at very high carbon prices.

4 Conclusion

The deployment of biomass for climate change mitigation cannot be guided by a single universal pathway. Our spatial analysis demonstrates that the economic viability of BES, BECCS, and BEBCS is dictated by a complex interplay of localized biophysical constraints and regional market dynamics.

While BECCS offers the highest theoretical carbon removal potential, its heavy infrastructure requirements and sensitivity to the cost of capital create significant barriers to entry, particularly where biomass is located far from suitable geological sinks. In contrast, biochar (BEBCS) circumvents these transport constraints and establishes a resilient economic niche in regions with degraded, low-CEC soils, where its agronomic co-benefits—such as liming equivalence and nutrient substitution—offset its lower electrical conversion efficiencies. Southern China exemplifies this dynamic, demonstrating that BEBCS is a robust, long-term mitigation strategy for specific agro-ecological zones rather than merely a transition technology.

Furthermore, baseline energy markets and agricultural policies heavily distort optimal pathways. In Europe, high wholesale electricity prices establish BES as the dominant pathway at low carbon prices, while in India, fertilizer subsidies artificially suppress biochar's agronomic value, pushing the market toward traditional BES.

Realizing the climate mitigation potential of global biomass requires moving beyond blanket technological mandates. Policymakers must adopt spatially explicit strategies that target BECCS deployment near geological sinks, incentivize BEBCS in agro-ecological zones suffering from soil degradation, and recognize the enduring role of BES in displacing highly carbon-intensive fossil grids.

5 Figures

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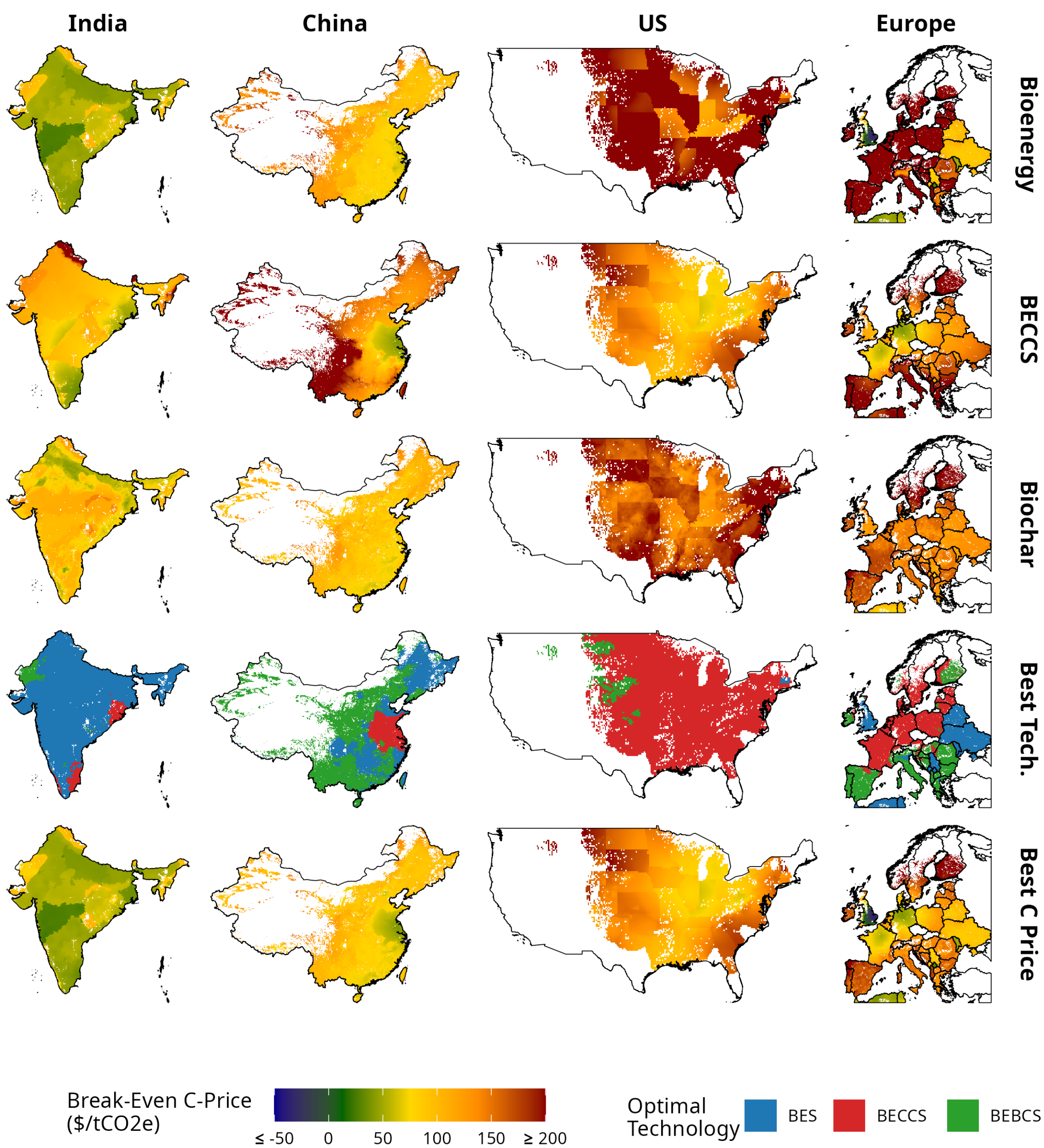

Figure 1: Minimum Break-even carbon price for each technology option. The technology with the lowest break-even price is shown as the "best technology" in the penultimate row. The final row shows the break-even price of this technology.

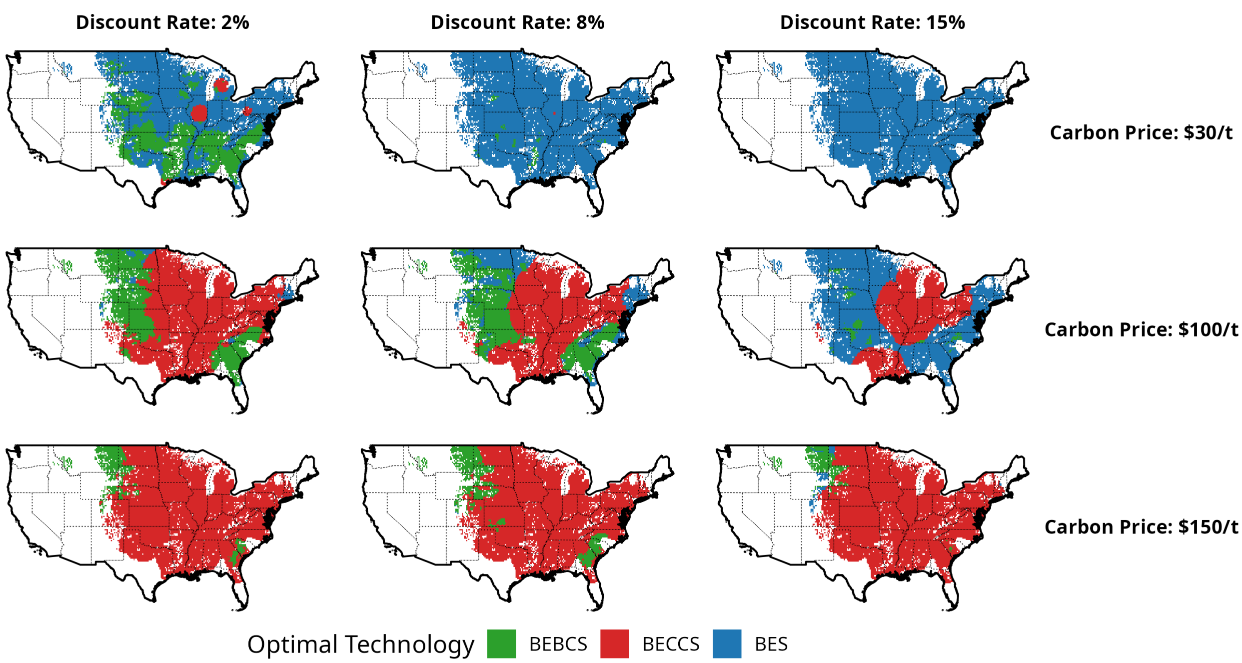

Figure 2: Optimal technology as a function of carbon price and discount rate (USA).

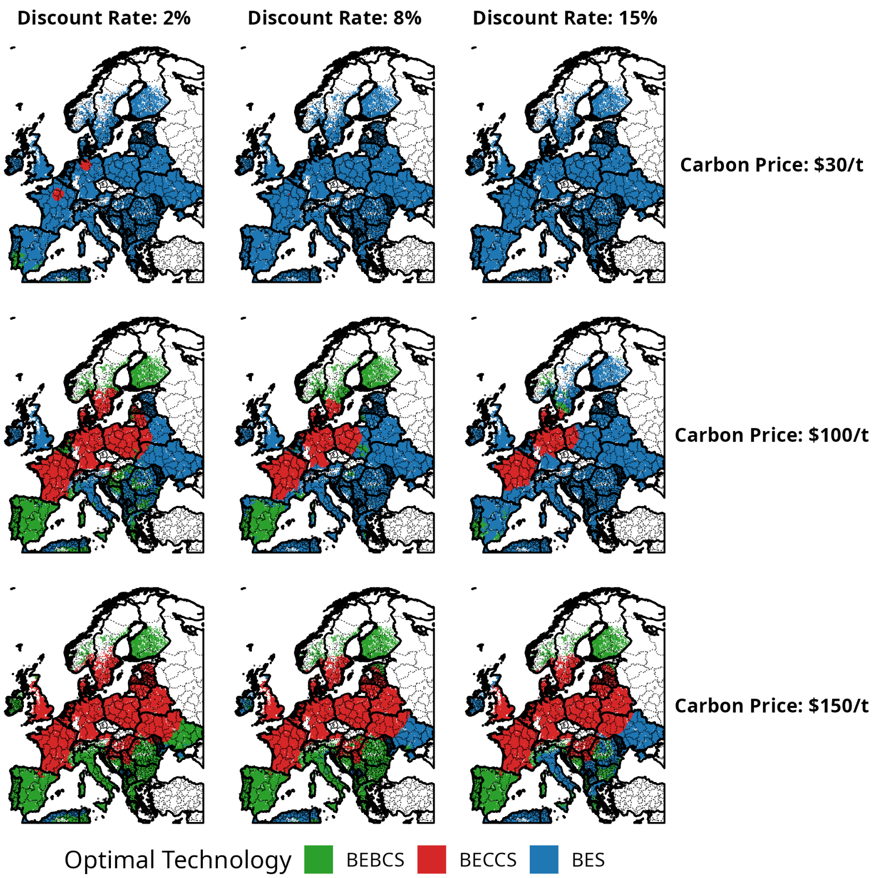

Figure 3: Optimal technology as a function of carbon price and discount rate (Europe).

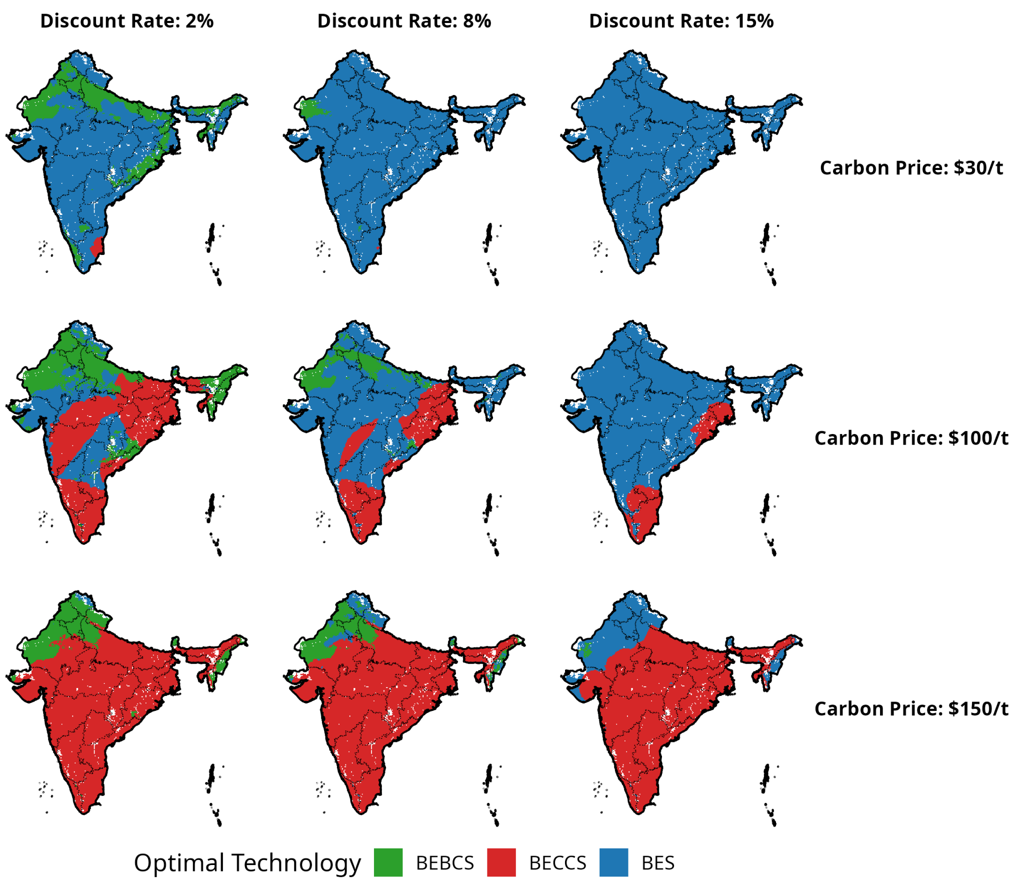

Figure 4: Optimal technology as a function of carbon price and discount rate (India).

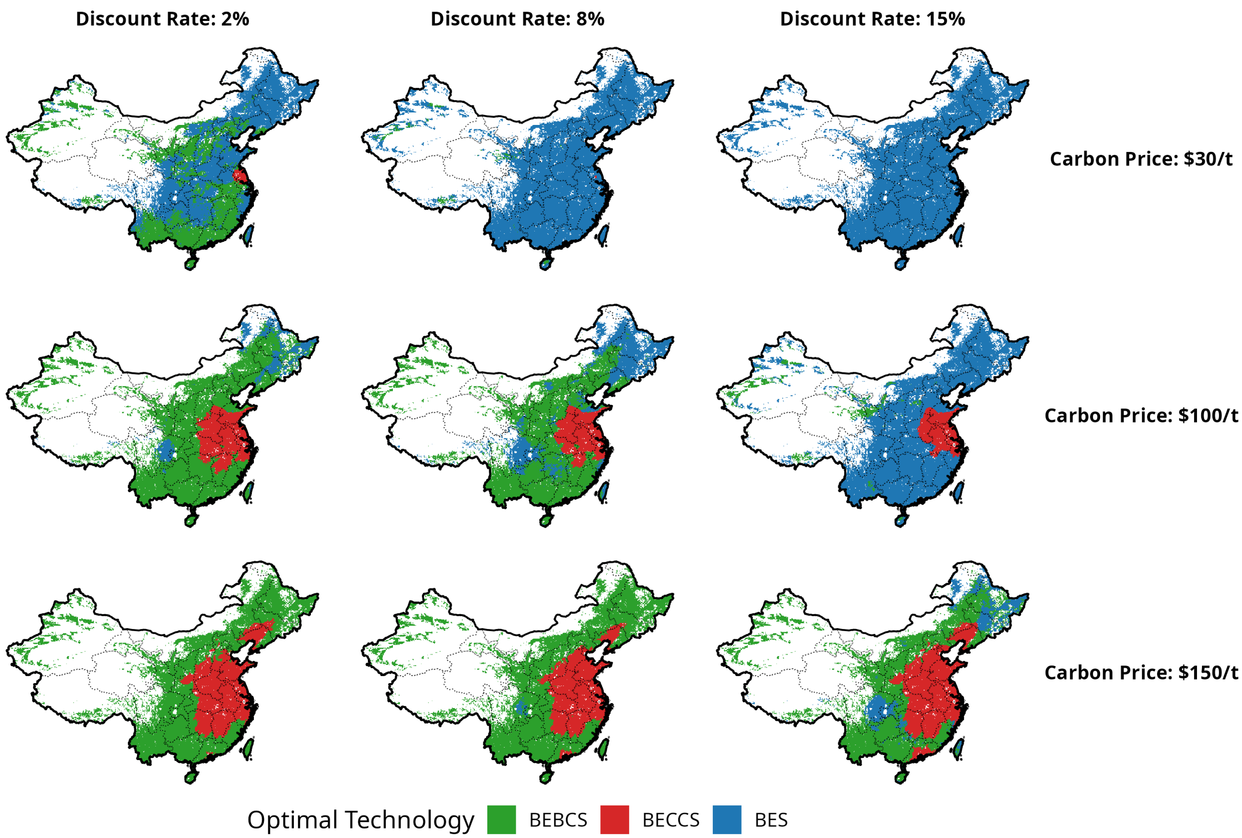

Figure 5: Optimal technology as a function of carbon price and discount rate (China).

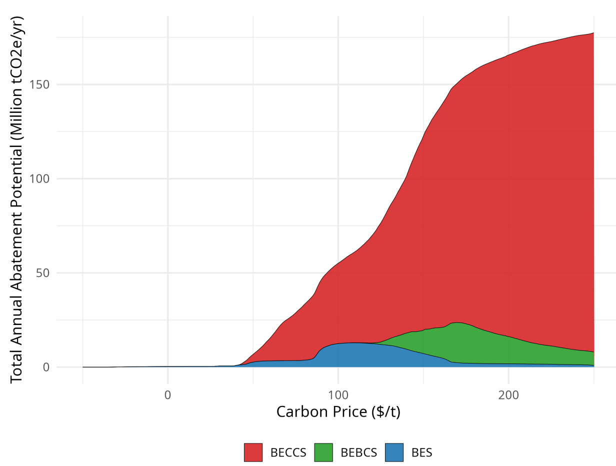

Figure 6: Marginal abatement cost curve for Europe.

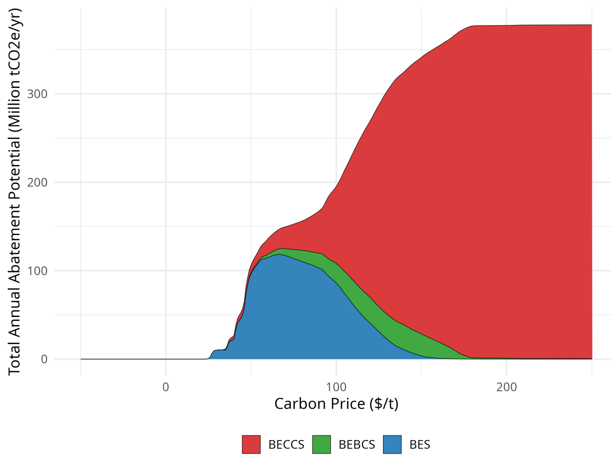

Figure 7: Marginal abatement cost curve for India.

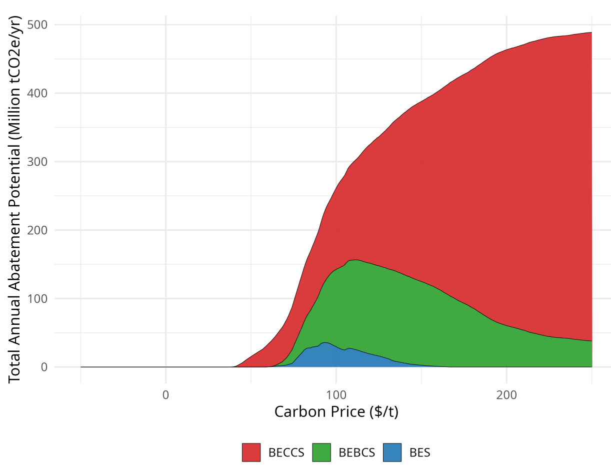

Figure 8: Marginal abatement cost curve for China.

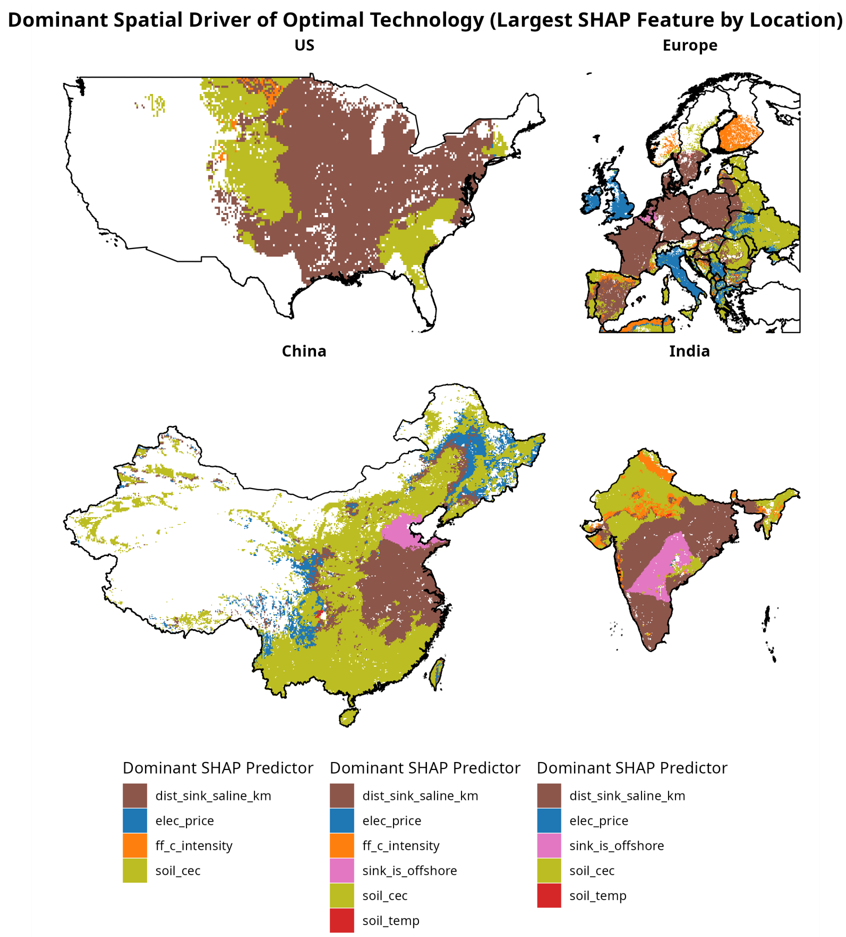

Figure 9: Dominant driver of breakeven carbon prices (Shapley decomposition).

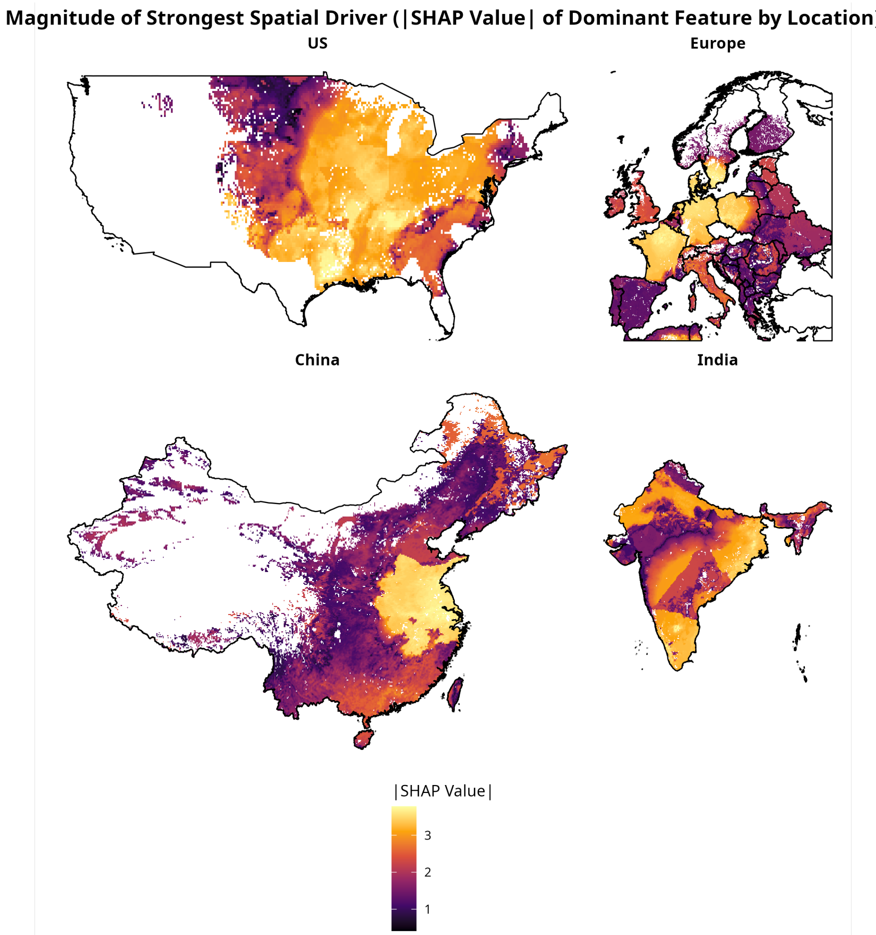

Figure 10: Magnitude of the dominant driver (Shapley decomposition).

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